OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2025–2026

Course Instance Codes	3BA1, 4BME1, 4BCT1, 1BMG1, 4BMS2, 4BS2, 4FM2, 1GDS1, 1MDX1
Exams	3rd Arts, 4th Science, Masters
Module Code	CS4423
Module	**Networks**
Paper No	1
External Examiner	Dr Michael Mackey
Internal Examiner	Dr Anton Baykalov ⋆
	Dr James Cruickshank
	Dr Tobias Rossmann
Instructions:	**Answer all 7 questions**
Duration	2 Hours
Number of Pages	4 (including this page)
Discipline	Mathematics

Requirements

Release in Exam Venue	No	☐	Yes	☑
Release to Library	No	☐	Yes	☑
Formulae and Tables	No	☐	Yes	☑
Handout	No	☑	Yes	☐
Cambridge Tables	No	☑	Yes	☐
Graph Paper	No	☑	Yes	☐
Log Graph Paper	No	☑	Yes	☐
Graphic material in colour	No	☑	Yes	☐

End of Requirements

Q1. [Total: 14 Marks] Let G_1 be a graph with adjacency matrix

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

(a) [4 Marks] Sketch the graph G_1 and label it in any way you want.

(b) [5 Marks] What are the *order* and the *size* of G_1? What are the order and the size of a *spanning tree* of G_1? Explain your answer.

(c) [2 Marks] Write down the degree sequence of G_1.

(d) [3 Marks] Does there exist a graph with degree sequence $(2, 3, 1, 2, 3)$? If so, give an example; if not, explain why.

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Q2. [Total: 15 Marks] Consider the *bipartite graph* G_2 with the set of nodes $\{0, 1, 2, 3, 4, 5\}$ and the set of edges $\{\{0, 3\}, \{1, 2\}, \{1, 4\}, \{1, 5\}, \{3, 4\}, \{3, 5\}\}$.

(a) [5 Marks] Sketch the graph and indicate a *two-colouring*.

(b) [5 Marks] Find an ordering of the nodes, with respect to which the adjacency matrix A of G_2 has the form

$$\begin{pmatrix} \mathbf{0} & C \\ C^T & \mathbf{0} \end{pmatrix}$$

where $\mathbf{0}$ denotes a block of zeros.

(c) [5 Marks] Let X and Y be the parts of G_2 corresponding to the two-colouring in (a) with $|X| > |Y|$. Construct the projection of G_2 onto X.

Q3. [Total: 15 Marks]

(a) [5 Marks] Let T_1 be a tree on the nodes $\{0, 1, 2, 3, 4, 5\}$ with degree sequence $(2, 1, 1, 3, 1, 2)$. How many distinct nodes appear in the *Prüfer code* of T_1? Explain your answer.

(b) [5 Marks] How does one construct the tree corresponding to a Prüfer code? Sketch the tree T_2 on the nodes $\{0, 1, \ldots, 7\}$ corresponding to the code $[3, 3, 3, 4, 4, 4]$.

(c) [5 Marks] Find a labeled tree T_3 (with the same set of labels $\{0, 1, \ldots, 7\}$) that is isomorphic but not equal to T_2 from part (b). Write down an isomorphism between T_2 and T_3.

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Q4. [Total: 14 Marks]

(a) [5 Marks] Describe *Breadth First Search* as an algorithm for computing distances between nodes in a (simple) graph. What is its input, what is its output, and what sequence of steps is taken to produce the output from the input?

(b) [9 Marks] Consider the graph G_3 shown in Figure 1. Show how to apply the Breadth First Search algorithm, starting at Node 0, to determine, for every node, its *predecessors* on the shortest path between it and Node 0.
Use this information to list all shortest paths from Node 0 to Node 7.

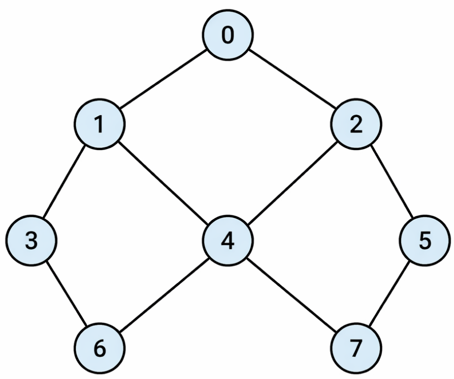

Figure 1: Graph G_3 from Question 4

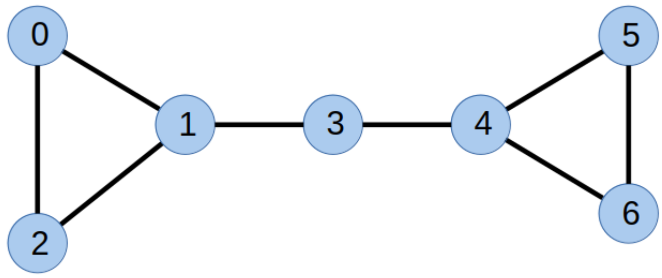

Figure 2: Graph G_4 from Question 5

Q5. [TOTAL: 14 MARKS] Consider the graph G_4 shown in Figure 2.

(a) [4 MARKS] Compute the degree and the *normalised degree centrality* for each node.

(b) [5 MARKS] Compute the distances from node 3 to all other nodes. Use these to compute the *closeness centrality* of node 3. Repeat the calculation for node 0. Which of the two nodes (0 or 3) is more central by this measure? Explain this intuitively.

(c) [5 MARKS] For node 3, list all pairs of nodes whose shortest paths pass through node 3. Compute the betweenness centrality of node 3.

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Q6. [TOTAL: 14 MARKS]

(a) [4 MARKS] Define the two *Erdös-Rényi* models $G_{ER}(n, m)$ and $G_{ER}(n, p)$ of random graphs. In these models, for what m and p, respectively, is the resulting graph empty (i.e. has no edges)?

(b) [5 MARKS] In each model, what is the probability that a randomly chosen graph G is the complete graph K_n? Justify your answer.

(c) [5 MARKS] In $G_{ER}(n, p)$, for what value of p is the expected number of edges equal to half the maximum possible number of edges?

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Q7. [TOTAL: 14 MARKS]

(a) [4 MARKS] Define the *characteristic path length* of a graph and the *small world behaviour*. What is the characteristic path length of $G_{ER}(n, m)$ approximately equal to?

(b) [5 MARKS] How does one compute the *transitivity* of a graph? Determine the graph transitivity of a random graph in the $G_{ER}(n, p)$ model.

(c) [5 MARKS] Describe the *Watts-Strogatz small-world model* (WS model). What properties does a random graph sampled from the WS model have, that one wouldn't find in a random graph sampled from the $G_{ER}(n, p)$ model, or in an (n, d)-circle graph?